A Field-Based Specification of Constraint Dynamics

1. Introduction

This note presents a condensed articulation of observable system structure implied by the Stress–Elasticity–Reaction (SER) framework.

The formal specification of SER, including its field-based formulation of stress, elasticity, leakage, and propagation, is developed separately (Castle, 2026).

The objective here is to outline the resulting dynamics, observable implications, and empirical correspondences that follow from that specification.

2. Field Interpretation

Let (i) index agents and (t) index time.

In the SER framework, stress is represented as the local realization of a distributed constraint field:
$S_{i,t} = S_t(i)$

Observable inputs ($X_{i,t}$) (cost burdens, price volatility, supply constraints, access limitations) condition this local realization but are not treated as independent causal sources.

The field representation should be interpreted as a reduced-form mapping of relational constraint discrepancy, rather than as a physical transmission field.

Absorption operates locally through buffering capacity, while leakage represents the portion of the field that exceeds local absorption and remains available for transmission:
$[S^{absorbed} i, t = h(E i, t, \theta_{i,t})]$
$[L_{i,t} = S_{i,t} - S_{i,t}^{absorbed}]$

Leakage therefore represents transmissible constraint within the field.

The full formal definition of these components is provided in the underlying specification; here they are presented to support interpretation of system behavior.

3. Elasticity and Reaction

The expressions below summarize the operative relationships of the SER framework. Detailed derivations and full structural assumptions are provided in the formal specification; the forms presented here are sufficient to illustrate system behavior and empirical correspondence.

Elasticity is a dynamic stock of buffering capacity:
$E_{i,t} = E_{i,t-1} + \Delta E_{i,t}, \ \Delta E_{i,t} = \alpha B_{i,t} - \beta L_{i,t}$

Elasticity and system efficiency act jointly as a local damping operator, reducing the amplitude of transmitted stress differentials:

$$d(E_{i,t}, \theta_{i,t}) = \frac{1}{1 + \kappa_E E_{i,t} + \kappa_\theta \theta_{i,t}}$$

Reaction reflects unresolved constraint:

$$R_{i,t} = a L_{i,t} - b E_{i,t} + \gamma R_{i,t-1} + e_{i,t}$$

A critical elasticity threshold (E_{crit}) separates regimes:

- ($E_{i,t} > E_{crit}$): damping
- ($E_{i,t} \leq E_{crit}$): amplification

Recovery exhibits hysteresis:

$$E_{recovery} > E_{decline}$$

4. System Conditions

Market variables are treated as observable expressions of constraint dispersion:

$$\Phi_t = \frac{D_t}{S_t^q}, \; S_t = f(P_t, \Phi_t, \sigma_P)$$

$$E_t = \gamma(RM_t, B_t, \theta_t)$$

These mappings describe realized conditions rather than primary causes. Stress may also accumulate through persistent structural imbalance independent of discrete events.

5. Propagation as Field Transport

To extend SER beyond local evaluation, stress is modeled as a distributed field evolving across connected nodes. These terms describe mathematical operators on the field representation and do not imply direct physical flow.

Stress motion:

$$m_{i,t} = S_{i,t} - S_{i,t-1}$$

Propagation may arise under sufficient gradient and constrained topology as directional flow from higher to lower field acceleration:

$$P_{i,t} = \lambda(R_t) \sum_j \tilde{w}_{ij} \, max(0; m_{j,t} - m_{i,t})$$

with

$$\widetilde{w_{ij,t}} = w_{ij} \; o_{ij,t} \; h_{ij,t} \; a_{j,t}$$

where:

- $o_{ij,t}$ = physical openness
- $h_{ij,t}$ = edge headroom / transmissive slack

- $a_{j,t}$ = destination acceptance factor

This corresponds to a **discrete gradient flow** on the stress field.

Formulation reflects the observed property that physically open paths may remain non-clearing when downstream acceptance is constrained, without $a_{j,t}$, the field paper cannot express receiver-limited transmission clean.

Transmission is damped locally:

$P^d_{i,t} = P_{i,t} \cdot d(E_{i,t}, \theta_{i,t})$

and extended with a time-based factor (aging)...

$P_{i,t} = P(S_{i,t}, L_{i,t}, \tau_{i,t})$

where $\tau_{i,t}$ captures concentration of burden over time.

A forecast decomposition may isolate the contribution of cross-node field transport:
$S^f_{i,t+1} = S_{i,t} + P^d_{i,t}$

Transmission intensity ($\lambda(R_t)$) varies with system regime, reflecting increased field coherence under elevated stress conditions.

$a_{j,t} = g(H_{j,t})$, $0 \le a_{j,t} \le 1$

where $H_{j,t}$ is local intake headroom or acceptance slack.

This factor captures the possibility that transmissible burden is limited not only by the sending node or edge state, but also by the receiving node's ability to absorb additional load

6. Alternative Propagation Forms

Different systems may emphasize different aspects of field structure.

Gradient form (persistent imbalance):
$P_{i,t} = \lambda(R_t) \sum_j w_{ij} max(0; S_{j,t} - S_{i,t})$

Blended form (transient + structural):
$P_{i,t} = \lambda(R_t) \sum_j w_{ij} [a\, max(0; m_{j,t} - m_{i,t}) + b\, max(0; S_{j,t} - S_{i,t})]$

Parameters (a) and (b) control the relative contribution of dynamic field motion and persistent gradients.

7. Leaked Stress Propagation

When buffering is significant, propagation is more accurately modeled using leaked stress:
$l_{i,t} = L_{i,t} - L_{i,t-1}$

$[P_{i,t}=\lambda(R_t)\sum_j w_{ij} max(0;l_{j,t}-l_{i,t})]$

Blended form:
$P_{i,t}=\lambda(R_t)\sum_j w_{ij}[a\,max(0;l_{j,t}-l_{i,t})+b\,max(0;L_{j,t}-L_{i,t})]$

Leakage represents the transmissible component of the field after local damping, ensuring propagation reflects unresolved constraint rather than total field intensity.

This representation follows directly from the underlying SER formulation and is presented here for clarity of interpretation.

8. Field Coherence and Empirical Interpretation

Empirical studies of commodity systems identify stronger cross-node connectedness under extreme conditions. (Bastianin, Andrea and Casoli, Chiara and Kocenda, Evzen and Li, Xiao, 2026)

Within the SER framework, this pattern is interpreted as increased field coherence, where stress becomes more synchronized and transmission more efficient as system constraint intensifies.

This provides an empirical correspondence between observed connectedness measures and the field-based dynamics described in the formal specification.

Observed decompositions into “common” and “idiosyncratic” components are treated as statistical projections of a single field at different scales. Interpreting them as distinct causal sources would introduce a separation incompatible with a field-based ontology.

Functional isolation may arise when physically open paths remain present but effective transmissibility collapses because downstream acceptance or local headroom is exhausted.

9. Recirculation and Persistence

Nodes may reintroduce stress into the system:
$[Q_{i,t}=\rho_i L_{i,t}+\omega_i P^d_{i,t}]$

Recirculation produces persistence and feedback, implying the field is not purely dissipative but may exhibit cumulative reinforcement through repeated transmission.

10. Observable System Structure

Aggregate system coupling may be defined as:
$[C_t=\frac{\sum i\neq j\, P_{ij,t}}{\sum_i S_{i,t}}]$

This provides an observable measure of field coherence, analogous to empirical connectedness metrics.

Nodes with persistently positive net transmission act as field emitters, while those with net absorption act as dampening regions, defining the functional structure of the system.

These measures correspond to empirical connectedness and centrality metrics, interpreted through the SER field framework rather than as independent structural primitives.

11. System Interpretation

Under the SER framework, system dynamics can be interpreted as arising from three interacting mechanisms:

- **Field dispersion**: spatial and temporal variation in constraint
- **Local damping**: elasticity and efficiency reducing transmission
- **Network transport**: propagation of leaked stress across nodes
- **Constraint-induced retention**: accumulation of unresolved discrepancy when transmissibility is limited

External conditions may perturb or reconfigure the field but do not fully determine its formation. Observed connectedness emerges from underlying field dynamics rather than discrete shock propagation.

These interpretations summarize the observable implications of the underlying SER formulation rather than introducing additional structural assumptions.

The complete formal specification of the SER framework, including derivation and full structural definition, is provided in Castle (2026).

Bastianin, Andrea and Casoli, Chiara and Kocenda, Evzen and Li, Xiao, Extreme Connectedness among Energy Transition Metals and Commodity Markets (April 03, 2026). FEEM Working Paper No. 13-2026, Available at SSRN: https://ssrn.com/abstract=6541178 or http://dx.doi.org/10.2139/ssrn.6541178

Castle, J. (2026). *A Formal Specification of the SER (Stress -Elasticity -Reaction) Framework Outcomes.* SSRN